

Lecture 9: Lattice Pricing & Dynamic Programming

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Based on lecture notes by me and many previous CS335 instructors

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Announcements:

- Midterm:
1. Wednesday
 2. Bring a
 3. Bring a
 4. Coverage.

Oct 28th at 4:30pm
non-programmable calculator.

Single-Sided 8.5x11 cheatsheet.
modules 1 to 5.

Formal Notation

V_N = expected partner wealth at step N (assuming optimal actions)

W_{N-1} = actual wealth of specific person $N - 1$

$$\bar{W}_{N-1} = E \left[W_{N-1} \mid \underbrace{W_{N-1} > V_N}_{\text{Stopping rule}} \right]$$

| means "conditional on"

= expected wealth of person $N - 1$, assuming you do stop at $N - 1$.

$$p_{N-1} = \text{prob}[W_{N-1} > V_N]$$

Example

Let's work backwards to find the expected wealth of your partner up to person $N - 2$. That is, find V_{N-2} .

Careful: This means the wealth of the person you *ultimately* marry, not necessarily the wealth of person $N - 2$.

Example

Expected partner wealth at step $N-1$, assuming you act optimally, on steps $(N-1, N)$:

$$V_{N-1} = p_{N-1} \bar{W}_{N-1} + (1 - p_{N-1}) V_N$$

Now $V_N = 0.5 = \frac{1+0}{2}$

$$p_{N-1} = \text{prob}(W_{N-1} > 0.5) = 0.5$$

End of Lecture 8

The example is not done!

Example

$$\overline{W}_{N-1} = E[W_{N-1} \mid W_{N-1} > 0.5] = \frac{0.5 + 1}{2} = 0.75$$

$$\therefore V_{N-1} = (0.5)(0.75) + (0.5)(0.5) = 0.625$$

Repeat the process:

$$V_{N-2} = P_{N-2} \cdot \overline{W}_{N-2} + (1 - P_{N-2}) V_{N-1}$$

$$V_{N-1} = 0.625$$

$$P_{N-2} = \text{prob}(W_{N-2} > 0.625) = 1 - 0.625 = 0.375$$

$$\begin{aligned} \overline{W}_{N-2} &= E[W_{N-2} \mid W_{N-2} > 0.625] = \frac{0.625 + 1}{2} \\ &= 0.8125 \end{aligned}$$

$$\therefore V_{N-2} = (0.375)(0.8125) + (0.625)(0.625) \\ \approx 0.695$$

Recall: V_{N-2} is the expected wealth of the person you end up marrying if you act optimally for $N-2, N-1, N$. Not the specific wealth of person $N-2$ (necessarily).

General Case

In general, at person k , we have the following rules:

$$V_k = p_k \bar{W}_k + (1 - p_k) V_{k+1}$$

$$p_k = \text{prob}[W_k > V_{k+1}]$$

$$\bar{W}_k = E[W_k \mid W_k > V_{k+1}]$$

We call W_k the Stopping Value - what you get if you stop/marry at k .

We call V_k the Continuation Value - *expected* value of continuing to play, i.e., move on to $k + 1$.

Optimal Strategy

We have a simple algorithm for the optimal strategy:

Algorithm Optimal Strategy

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1: if stopping value > continuation value then  
2:   stop  
3: else  
4:   continue  
5: end if
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Marriage Problem - Outcome

In our problem...

$$p_k = 1 - V_{k+1}$$

$$\bar{W}_k = \frac{V_{k+1} + 1}{2}$$

$$\begin{aligned} V_k &= p_k \bar{W}_k + (1 - p_k) V_{k+1} \\ &= (1 - V_{k+1}) \left(\frac{1 + V_{k+1}}{2} \right) + V_{k+1}^2 \end{aligned}$$

For the $N = 10$ partners case, $V_1 \approx 0.861$.

Convergence of a numerical scheme

Definition: A numerical method is p^{th} order accurate if the error is $O(\Delta t^p)$, where Δt is the time step size.

That is, $V_{\Delta t}(S, t) = V_{\text{exact}}(S, t) + C \Delta t^p + O(\Delta t^{p+1})$ for some C independent of Δt .

Effectively, this describes how rapidly we converge to the true solution, as we take smaller time steps.

1st order accurate: cut Δt by half, error is about 1/2 as large.

2nd order accurate: cut Δt by half, error is about 1/4 as large.

Estimating order of accuracy

How to determine p numerically? (Especially if we don't *know* that exact solution??)

Consider error ratios for different *finite* timesteps sizes.

$$V_{\Delta t}(S, t) - V_{\frac{\Delta t}{2}}(S, t) = \left[\cancel{V_{\text{exact}}(S, t)} + C \Delta t^p + O(\Delta t^{p+1}) \right] \\ - \left[\cancel{V_{\text{exact}}(S, t)} + C \left(\frac{\Delta t}{2} \right)^p + O\left(\left(\frac{\Delta t}{2} \right)^{p+1} \right) \right]$$

or

$$V_{\Delta t}(S, t) - V_{\frac{\Delta t}{2}}(S, t) \\ = C \Delta t^p - C \left(\frac{\Delta t}{2} \right)^p + O(\Delta t^{p+1})$$

Estimating order of accuracy

So

$$V_{\Delta t}(S, t) - V_{\frac{\Delta t}{2}}(S, t) = C\Delta t^p - \frac{C}{2^p}\Delta t^p + O(\Delta t^{p+1})$$

and likewise,

$$V_{\frac{\Delta t}{2}}(S, t) - V_{\frac{\Delta t}{4}}(S, t) = \frac{C}{2^p}\Delta t^p - \frac{C}{4^p}\Delta t^p + O(\Delta t^{p+1})$$

Ratios of error differences are:

$$\begin{aligned} \frac{V_{\Delta t}(S, t) - V_{\frac{\Delta t}{2}}(S, t)}{V_{\frac{\Delta t}{2}}(S, t) - V_{\frac{\Delta t}{4}}(S, t)} &= \frac{C(\Delta t)^p \left(1 - \frac{1}{2^p}\right) + O(\Delta t^{p+1})}{C(\Delta t)^p \left(\frac{1}{2^p} - \frac{1}{2^{2p}}\right) + O(\Delta t^{p+1})} \\ &= 2^p + O(\Delta t) \end{aligned}$$

Estimating order of accuracy

So, as we (re)compute solutions for smaller timesteps Δt , these ratios of error differences approach 2^p (in the limit).

e.g. if the ratio approaches $\approx 2^3 = 8$, then the numerical method is 3^{rd} order accurate.

Inconveniently, this estimate (and our lattice option pricing approach!) is only accurate/convergent *in general* if V is sufficiently *smooth*.

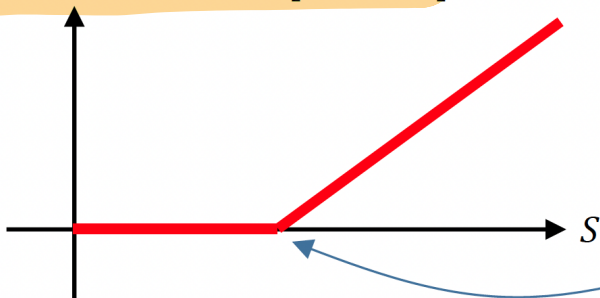
Why inaccurate?

Our derivation of the no arbitrage lattice depended on approximating a derivative (the option delta) with finite differences. Recall:

$$\delta_j^n = \frac{V_{j+1}^{n+1} - V_j^{n+1}}{S_{j+1}^{n+1} - S_j^{n+1}} \approx \frac{\partial V}{\partial S}$$

However, the payoff value is non-smooth due to the max function.

payoff = $\max(S - K, 0)$ [for a call]



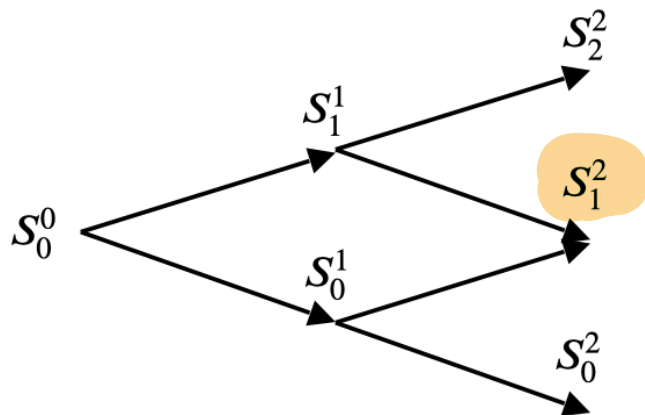
That's a sharp kink
(discontinuity in the
derivative at strike K).

A special case

A slope/derivative estimate using two points (with one on either side of the strike/kink) will not be accurate!

However, the ratio estimate does still remain valid *if* there is a lattice node *exactly* at the strike value K at final time T .

e.g., if N is even, and $S_0^0 = K$.
In this figure, $N = 2$, so $S_0^0 = S_1^2$.
Thus if $S_0^0 = K$, then $S_1^2 = K$,
and this estimate will work.



Smoothing the payoff

This case doesn't occur in general, but we can instead *smooth* the payoff ourselves to recover convergence. i.e., construct a new payoff to lessen/eliminate the kink.

We'll again consider our no-arb lattice with $N = \frac{T}{\Delta t}$ steps.

Stock value satisfies: $S_j^N = S_0^0 e^{(2j-N)\sigma\sqrt{\Delta t}}$.

Variable X is defined as a function of S : $X_j^N = \log(S_j^N)$.

Within a column of the tree, nodes are spaced by $2\sigma\sqrt{\Delta t}$. i.e.,

$$X_{j+1}^N = X_j^N + 2\sigma\sqrt{\Delta t}$$

$$X_{j-1}^N = X_j^N - 2\sigma\sqrt{\Delta t}$$

Smoothing the payoff

Our strategy is to replace the discrete payoff values at nodes with their integrated averages over intervals surrounding them.

Let $g(X)$ = regular option payoff (of the chosen type). e.g.,

$$g(X) = \begin{cases} \max(e^X - K, 0) = \max(S - K, 0) & \text{[call],} \\ \max(K - e^X, 0) = \max(K - S, 0) & \text{[put].} \end{cases}$$

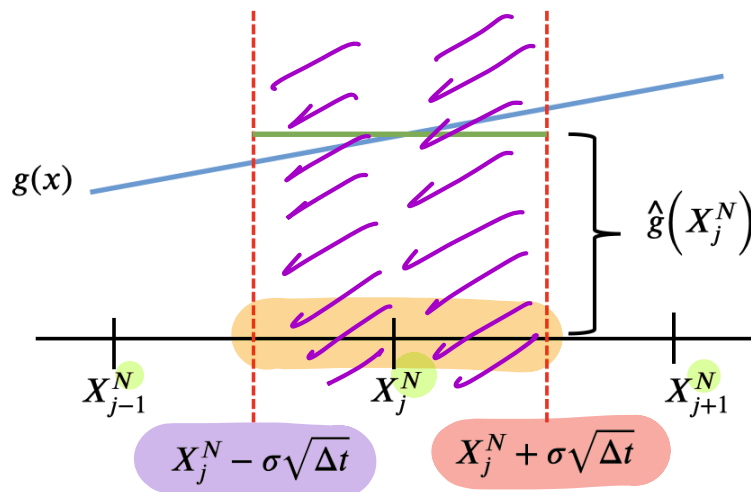
Define: $\hat{g}(X_j^N)$ = averaged value around node X_j^N . i.e., the smoothed payoff.

So, determine $\hat{g}(X_j^N)$ for all j , and use to replace the lattice payoffs, $g(X_j^N)$

Smoothing the payoff

Mathematically, we write this as:

$$\hat{g}(X_j^N) = \frac{1}{2\sigma\sqrt{\Delta t}} \int_{X_j^N - \sigma\sqrt{\Delta t}}^{X_j^N + \sigma\sqrt{\Delta t}} g(x) dx$$



Each integration interval extends halfway to the next node on either side, so the length (between the dashed red lines) is $2\sigma\sqrt{\Delta t}$.

Smoothing the payoff

There are three cases to consider:

- 1 Intervals above the strike (kink).
- 2 Intervals below the strike.
- 3 Intervals that *cross* the strike.

Let's work it out...

Smoothing the payoff

$$\hat{g}(X_j^N) = \frac{1}{2\sigma\sqrt{\Delta t}} \int_{X_j^N - \sigma\sqrt{\Delta t}}^{X_j^N + \sigma\sqrt{\Delta t}} g(x) dx$$

Let $u = x - X_j^N$ Then

$$\hat{g}(X_j^N) = \frac{1}{2\sigma\sqrt{\Delta t}} \int_{-\sigma\sqrt{\Delta t}}^{\sigma\sqrt{\Delta t}} g(u + X_j^N) du.$$

For a put option:

$$\hat{g}(X_j^N) = \frac{1}{2\sigma\sqrt{\Delta t}} \int_{-\sigma\sqrt{\Delta t}}^{\sigma\sqrt{\Delta t}} \max(k - e^{X_j^N + u}, 0) du$$

Smoothing the payoff

$$= \frac{1}{2\sigma\sqrt{\Delta t}} \int_{-\sigma\sqrt{\Delta t}}^{\sigma\sqrt{\Delta t}} \max(K - S_j^N e^u, 0) du$$

Case 1: Above the kink/strike

If $S_j^N e^{-\sigma\sqrt{\Delta t}} > K$ (i.e., at $u = -\sigma\sqrt{\Delta t}$)

Then $\max(K - S_j^N e^{-\sigma\sqrt{\Delta t}}, 0) = 0 \quad \forall u \in [-\sigma\sqrt{\Delta t}, \sigma\sqrt{\Delta t}]$

Hence $\hat{g}(X_j^N) = 0$

Case 2: Below the strike.

If $S_j^N e^{\sigma\sqrt{\Delta t}} < K$ (i.e., when $u = \sigma\sqrt{\Delta t}$)

Smoothing the payoff

Then $\max(k - S_j^N e^u, 0) = k - S_j^N e^u, \forall u \in [-\sigma\sqrt{\Delta t}, \sigma\sqrt{\Delta t}]$

Hence
$$\hat{g}(X_j^N) = \frac{1}{2\sigma\sqrt{\Delta t}} \int_{-\sigma\sqrt{\Delta t}}^{\sigma\sqrt{\Delta t}} (k - S_j^N e^u) du$$

$$= k - S_j^N \left[\frac{e^{\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}{2\sigma\sqrt{\Delta t}} \right]$$

Case 3: Crossing the strike.

$$S_j^N e^{-\sigma\sqrt{\Delta t}} \leq k \leq S_j^N e^{\sigma\sqrt{\Delta t}}, \text{ then find } u \text{ at}$$

the strike:

$$S_j^N e^u = K \implies u = \log\left(\frac{K}{S_j^N}\right)$$

only need to integrate over the non-zero range,
i.e., from end point to the strike at

$$u = \log\left(\frac{K}{S_j^N}\right)$$

$$\hat{g}(X_j^N) = \frac{1}{2\sigma\sqrt{\Delta t}} \int_{-\sigma\sqrt{\Delta t}}^{\log(K/S_j^N)} (K - S_j^N e^u) du$$

$$= \frac{1}{2\sigma\sqrt{\Delta t}} \left[[Ku]_{-\sigma\sqrt{\Delta t}}^{\log(K/S_j^N)} \right.$$

$$\left. - S_j^N [e^u]_{-\sigma\sqrt{\Delta t}}^{\log(K/S_j^N)} \right]$$

$$= \frac{1}{2\sigma\sqrt{\Delta t}} \left(K(\log(K/S_j^N) + \sigma\sqrt{\Delta t}) \right.$$

$$\left. - S_j^N \left(\frac{K}{S_j^N} - e^{-\sigma^2 \Delta t} \right) \right)$$

End of Lecture 9